

Experiment - 5

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Subject Name: Machine Learning Lab

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Q. Evaluate Machine learning algorithms to real-time data.

Aim/Overview of the practical:

- ❖ Write a program based on Artificial neural. Also specify the data set used with number of input and output variables.

Task to be Done:

- Implement a simple **Artificial Neural Network (ANN)** model using Python and NumPy.
- Use the **Iris dataset** from sklearn.datasets as the real-world dataset.
- Preprocess the dataset by performing **one-hot encoding** on target labels and **standardization** on feature values.
- Define network parameters such as **input size (4)**, **hidden layer size (5)**, and **output size (3)**.
- Implement **forward propagation** using **ReLU** and **Softmax** activation functions.
- Calculate and display the **initial accuracy** of the network before any training.

Code for Experiment

```
import numpy as np
from sklearn.datasets import load_iris
from sklearn.preprocessing import OneHotEncoder, StandardScaler

data = load_iris()
X = data.data
y = data.target.reshape(-1, 1)

encoder = OneHotEncoder(sparse_output=False)
y_encoded = encoder.fit_transform(y)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

np.random.seed(1)
input_size = 4
hidden_size = 5
output_size = 3

W1 = np.random.randn(input_size, hidden_size)
b1 = np.zeros((1, hidden_size))
W2 = np.random.randn(hidden_size, output_size)
b2 = np.zeros((1, output_size))

def relu(x):
    return np.maximum(0, x)

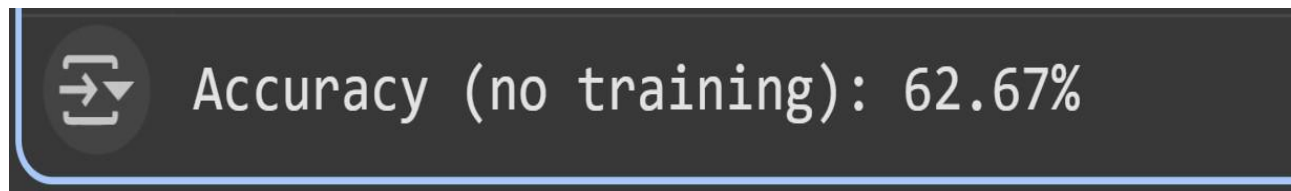
def softmax(z):
    exp_z = np.exp(z - np.max(z, axis=1, keepdims=True))
    return exp_z / np.sum(exp_z, axis=1, keepdims=True)

def forward(X):
    Z1 = np.dot(X, W1) + b1
    A1 = relu(Z1)
    Z2 = np.dot(A1, W2) + b2
    A2 = softmax(Z2)
    return A2
```

```
y_pred = forward(X_scaled)
predicted_classes = np.argmax(y_pred, axis=1)
actual_classes = np.argmax(y_encoded, axis=1)

accuracy = np.mean(predicted_classes == actual_classes)
print(f'Accuracy (no training): {accuracy*100:.2f}%')
```

Output :-



Accuracy (no training): 62.67%

Learning Outcomes (What I Have Learnt):

- Understood how an **Artificial Neural Network** processes data through layers of neurons.
- Learned the concept of **input, hidden, and output layers** and how weights and biases are initialized.
- Gained knowledge of **activation functions** like ReLU (for hidden layers) and Softmax (for classification output).
- Learned how to apply **data preprocessing** techniques such as **standardization** and **one-hot encoding**.
- Observed how the ANN makes predictions even before training, emphasizing the need for **training and optimization**.
- Enhanced understanding of how **ANNs classify real-world data** (in this case, different species of flowers) based on feature patterns.